PO11Q - Quantitative Political Analysis:
From Measurement to Inference

Formulae Collection and Notation

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Table 1: Formulae for PO11Q

Statistic	Formula
Confidence Interval	$Pr(\bar{y} - t_{\alpha/2} \cdot se \leq \mu \leq \bar{y} + t_{\alpha/2} \cdot se) = 1 - \alpha$
Deviation	$d = y_i - \bar{y}$
Mean	$\bar{y} = \frac{\Sigma y_i}{n}$
	$\mu = \Sigma y P(y) = E[y]$
Position of pth percentile	$P = (n+1) \cdot \frac{p}{100}$
Range	$y_{range} = y_{max} - y_{min}$
Standard Deviation	$s = \sqrt{\frac{\Sigma(y_i - \bar{y})^2}{n-1}}$
Standard Error	$\sigma_{\bar{y}} = \frac{\sigma}{\sqrt{n}}$
	$se = \frac{s}{\sqrt{n}}$
t-test	$t = \frac{\bar{y} - \mu_0}{se}$
Variance	$s^2 = \frac{\Sigma(y_i - \bar{y})^2}{n-1}$
z-score	$z = \frac{y - \mu}{\sigma}$
Cohen's *d* (Observed)	$d_{\text{obs}} = \frac{\bar{y} - \mu_0}{s}$
Cohen's *d* (True)	$d_{\text{true}} = \frac{\mu - \mu_0}{\sigma}$
Noncentrality Parameter	$\lambda = \frac{\mu - \mu_0}{\sigma_{\bar{y}}}$
	$\lambda = d_{\text{test}}\sqrt{n}$
Statistical Power	Power = P(reject H_0 | H_A is true)
	$1 - \beta$

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Table 2: Notation for PO11Q

Symbol	Explanation
d	Deviation of an observation y_i from the sample mean $\bar{y}$ – the difference between them, $d = y_i - \bar{y}$.
n	Sample size; the number of observations in the sample.
s	Sample standard deviation; the square root of the variance, $s = \sqrt{\frac{\Sigma(y_i-\bar{y})^2}{n-1}}$.
s^2	Sample variance; the squared standard deviation, $s^2 = \frac{\Sigma(y_i-\bar{y})^2}{n-1}$.
$\bar{y}$	Sample mean; the sum of the observations divided by their number, $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$.
y_i	The i-th observation in the sample, where $i = 1, 2, \ldots, n$.
f	Absolute frequency; the number of times a particular value or category occurs in the data.
cf	Cumulative absolute frequency; the running total of absolute frequencies up to and including a given value.
rf	Relative frequency; the proportion of observations taking a particular value, $rf = f/n$.
crf	Cumulative relative frequency; the running total of relative frequencies up to and including a given value, ranging from 0 to 1.
E[x]	Expected value of x; the long-run average of a random variable over infinitely many repetitions, i.e. the mean of its probability distribution.
μ	Population mean of the variable of interest; a parameter – a fixed value characterising the population – estimated by the sample mean $\bar{y}$.
se	Estimated standard error; the standard deviation of the sampling distribution of $\bar{y}$ estimated from the sample, $se = \frac{s}{\sqrt{n}}$ (replacing the unknown σ with s).
σ	Population standard deviation; the typical distance of individual values from the population mean μ. Usually unknown and estimated by s.
$\sigma_{\bar{y}}$	True standard error; the standard deviation of the sampling distribution of $\bar{y}$, $\sigma_{\bar{y}} = \frac{\sigma}{\sqrt{n}}$ (uses the known population σ).
t_{crit}	Threshold value from the t-distribution with $n - 1$ degrees of freedom that defines the boundaries of a confidence interval; for a $(1 - \alpha)$ interval, $P(-t_{\text{crit}} \leq t \leq t_{\text{crit}}) = 1 - \alpha$.
t_{obs}	Standardised test statistic that measures how many standard errors a sample statistic (typically a sample mean) lies from a hypothesised population parameter.
z	z-score (also z-value); expresses in standard-deviation units how far an observation falls from the mean, $z = \frac{y-\mu}{\sigma}$.
N	Population size; the number of observations in the population.
df	Degrees of freedom, express constraints on our estimation process
t	t-score, see t_{crit} and t_{obs}
$P()$	Probability of a statement
α	Significance level; the probability of rejecting a true null hypothesis (a Type I error). Common levels are 0.05, 0.01, or 0.001; it also fixes the confidence level $1 - \alpha$.
β	Probability of a Type II error (a false negative) – not rejecting a null hypothesis that is actually false. Determined by the effect size, sample size, and significance level.
H_0	Null Hypothesis, typically asserting that there is no effect or no difference.
H_1	Alternative Hypothesis, proposes that there is an effect or a difference
μ_0	Null Hypothesis Value

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Table 2 – continued

Symbol	Explanation
p	p-value, indicates the probability of obtaining a result equal to, or even more extreme than the observed value in the direction of the alternative hypothesis, assuming the null hypothesis is true.
$1 - \beta$	Statistical power; the probability of detecting a true effect, i.e. rejecting H_0 when it is false. Grows with larger effects and larger samples.
d_{obs}	Observed, standardised effect size (Cohen's d); the magnitude of the difference, expressed in standard-deviation units. Larger effects are easier to detect.
d_{test}	Assumed effect size, for example in a power or sample-size calculation; the value that enters the noncentrality parameter, $\lambda = d_{\text{test}} \cdot \sqrt{n}$.
d_{true}	The true effect size in the population, which the observed effect d_{obs} estimates.
λ	Noncentrality parameter; how many standard errors the true effect lies from the null value, $\lambda = d_{\text{test}} \cdot \sqrt{n}$; it shifts the non-central t-distribution under the alternative hypothesis.